

Extension to non-front-parallel placement.

The front-parallel case above is a special case of a more general result. We now show that the key linear conclusion still holds when the cube is placed with a fixed yaw angle θ , i.e., when its front face is not necessarily parallel to the image plane.

Let the cube center in the camera coordinate system be $\mathbf{x}^c = [x^c, y^c, z^c]^\top$, with side length d . We consider a pure yaw rotation around the vertical axis, and define the rotated local basis vectors in the camera coordinate system as

$$\mathbf{e}_x'^c(\theta) = \begin{bmatrix} \cos \theta \\ 0 \\ -\sin \theta \end{bmatrix}, \quad \mathbf{e}_z'^c(\theta) = \begin{bmatrix} \sin \theta \\ 0 \\ \cos \theta \end{bmatrix}. \quad (1)$$

Here, $\mathbf{e}_x'^c(\theta)$ is the outward normal direction of the lateral faces, and $\mathbf{e}_z'^c(\theta)$ is the in-plane horizontal direction of the lateral face. Let $\sigma \in \{+1, -1\}$ indicate which lateral face is visible, where $\sigma = +1$ denotes one side face and $\sigma = -1$ denotes the opposite side face. Then the visible lateral face can be parameterized as

$$\mathbf{p}^c(s, t) = \mathbf{x}^c + \sigma \frac{d}{2} \mathbf{e}_x'^c(\theta) + s \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + t \mathbf{e}_z'^c(\theta), \quad s, t \in \left[-\frac{d}{2}, \frac{d}{2}\right]. \quad (2)$$

The two opposite vertical edges of this face correspond to $t = \pm \frac{d}{2}$. Their x - and z -coordinates in the camera coordinate system are

$$x_\pm^c = x^c + \frac{d}{2}(\sigma \cos \theta \pm \sin \theta), \quad z_\pm^c = z^c + \frac{d}{2}(-\sigma \sin \theta \pm \cos \theta). \quad (3)$$

Under the undistorted pinhole model, the projected heights of these two vertical edges are

$$h_\pm = \frac{f_y d}{z_\pm^c}. \quad (4)$$

The horizontal separation between the two projected edges is

$$\Delta u_\sigma(\theta) = f_x \left| \frac{x_+^c}{z_+^c} - \frac{x_-^c}{z_-^c} \right|. \quad (5)$$

Substituting the expressions of $x_\pm^c$ and $z_\pm^c$ and simplifying gives

$$\Delta u_\sigma(\theta) = \frac{f_x}{z_+^c z_-^c} \left| -d \cos \theta x^c + d \sin \theta z^c - \sigma \frac{d^2}{2} \right|. \quad (6)$$

Importantly, for fixed z^c , d , and θ , both z_+^c and z_-^c are independent of x^c . Hence, $\Delta u_\sigma(\theta)$ is an affine function of x^c inside the absolute value.

Approximating the projected quadrilateral by a trapezoid, the image-plane area of the visible lateral face is

$$a_{\text{px}}^{(\sigma, \theta)} = \frac{1}{2} (h_+ + h_-) \Delta u_\sigma(\theta). \quad (7)$$

Since h_+ and h_- are also independent of x^c when z^c and θ are fixed, we obtain

$$a_{\text{px}}^{(\sigma, \theta)} = \kappa_\sigma(\theta, z^c) \left| -d \cos \theta x^c + d \sin \theta z^c - \sigma \frac{d^2}{2} \right|, \quad (8)$$

where $\kappa_\sigma(\theta, z^c) > 0$ is a constant with respect to x^c .

Therefore, under a fixed yaw angle θ and a fixed visible-face branch σ , the projected side-face area remains a piecewise affine function of the lateral position x^c . When $\cos \theta \neq 0$, Eq. (8) can be equivalently written as

$$a_{\text{px}}^{(\sigma, \theta)} = \lambda_\sigma(\theta, z^c) |x^c - \beta_\sigma(\theta, z^c)|, \quad (9)$$

for some constants $\lambda_\sigma(\theta, z^c) > 0$ and $\beta_\sigma(\theta, z^c)$. When $\cos \theta = 0$, the dependence on x^c degenerates to a constant, which is still an affine special case. This shows that the front-parallel derivation is not a restrictive requirement; it is simply the $\theta = 0$ special case of a more general piecewise affine relationship.

This conclusion is valid under the following conditions: (i) the undistorted pinhole camera model holds; (ii) the yaw angle θ is fixed; (iii) the same lateral face remains visible, i.e., no visibility switching occurs; and (iv) all relevant face vertices satisfy $z_\pm^c > 0$, so that the face remains in front of the camera.